Aerodynamic Coefficients of a Coupled Wing–Body Configuration Across the Separation Boundary — Part I: Attachment Lines, the Separation March, and the Attached-Flow Envelope

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Coupling a boundary-layer method to a potential-flow solver requires knowing two things before the march can begin: where the flow attaches, and what the edge-velocity distribution is measured from that point. This paper presents a method for extracting both from a coupled wing–body panel solve in which the lifting surfaces carry horseshoe vortices and the fuselage carries constant-strength source panels in one blocked linear system. The two halves of the airframe require different treatments, and the difference is structural rather than incidental. A source-panelled body is a real closed surface whose stagnation points exist in the solution and are located by searching it; the skin-flow topology is resolved in the panelling's own surface-index space, critical points are found as common zeros of the

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contravariant velocity components, and they are classified from a strain-rate tensor evaluated in a local orthonormal frame. A vortex lattice, by contrast, is a zero-thickness camber sheet on which the flow never stops, and no search of its field can produce an attachment line. The lattice instead supplies the local induced flow to a two-dimensional Hess–Smith solve of the section's reconstructed thick contour, posed in the plane normal to the leading edge rather than streamwise. That choice is shown to be load-bearing under sideslip: the effective sweep moves in opposite directions on the two wings, so the attachment line and its Reynolds number become asymmetric, and a streamwise formulation returns a symmetric answer to an asymmetric question. The method is verified against closed-form results — exact stagnation-point locations and strain rates on a sphere across the full attitude range, exact surface velocity on a circular cylinder, and the $\sqrt{r_{LE}}$ scaling of stagnation-point movement predicted by matched asymptotics — and yields, per spanwise station, the attachment-point location, the Thwaites stagnation momentum thickness that starts a chordwise march, and Poll's attachment-line Reynolds number.

Nomenclature

$\mathbf{a}_i, \mathbf{a}_j$	covariant surface basis vectors, m
b	wing span, m
c	streamwise chord, m
c_n	chord in the leading-edge-normal plane, m
C_p	pressure coefficient
C_L, C_{D_i}, C_Y	lift, induced-drag and side-force coefficients
C_D, C_{D_0}	total and parasite drag coefficients, $C_D = C_{D_i} + C_{D_0}$
C_l, C_m, C_n	rolling, pitching and yawing moment coefficients
h	streamline divergence (spreading) metric
H	boundary-layer shape factor
J	surface strain-rate tensor, s^{-1}
L	body length, m
$\hat{\mathbf{n}}$	unit outward surface normal
$\bar{R}$	attachment-line Reynolds number, $W_e \eta / \nu$
r_{LE}	leading-edge radius, m
s	surface arc length, m
u, v	contravariant surface velocity components, s^{-1}
U_e	boundary-layer edge velocity, m/s

$\mathbf{V}_t$	surface-tangential velocity, m/s
W_e	velocity component along the leading edge, m/s
α	angle of attack, deg
α_n	incidence in the leading-edge-normal plane, deg
β	sideslip angle, deg
Γ	circulation, m^2/s
η	attachment-line length scale, $\sqrt{\nu/(dU_e/ds)}$, m
θ	momentum thickness, m
λ	Thwaites' pressure-gradient parameter
Λ	leading-edge sweep angle, deg
Λ_{eff}	sweep relative to the local flow, deg
ν	kinematic viscosity, m^2/s
σ	source-panel strength, m/s
ψ, ζ	section coordinates normalized by chord

I. Introduction

Panel methods remain the workhorse of configuration-level aerodynamic analysis because they resolve the inviscid field of a complete airframe in seconds per operating point, which is what makes attitude sweeps, stability-derivative tables and design optimization tractable. What they do not resolve is the boundary layer, and the standard remedy — coupling an integral boundary-layer method to the inviscid solution[1,2] — has a prerequisite that is easy to state and less easy to satisfy on a three-dimensional configuration. Every integral method marches along an external streamline from an attachment point. Before it can run, the attachment point must be located and the edge velocity must be known as a function of arc length measured from it.

On a two-dimensional airfoil this is routine: a panel method on the real contour gives the surface velocity, whose sign change near the nose is the stagnation point. On a three-dimensional airframe analyzed by a vortex lattice it is not routine at all, for a reason that is worth making explicit because it is frequently elided. A vortex lattice is not a model of a surface. It is a model of a camber sheet of zero thickness, on which the flow does not stop anywhere; the leading edge of such a sheet carries a square-root velocity singularity rather than a stagnation point. Any procedure that claims to locate a wing's attachment line by searching the three-dimensional lattice field is therefore reporting a property of its own discretization. The information required — the leading-edge radius and the surface curvature near it — was discarded when the section was reduced to its mean line.

A source-panelled body is in the opposite position. It is a genuine closed surface carrying a genuine velocity field, and its stagnation points exist in the solution as zeros of the surface-tangential velocity. They can be found rather than modeled, together with the whole skin-flow topology: attachment and separation nodes, saddles whose separatrices divide the surface, and foci marking vortical liftoff.[3,4]

This paper presents a treatment that respects that asymmetry rather than papering over it, implemented in an open-source coupled wing–body– propeller solver.[5] Section II summarizes the coupled vortex/source system and the velocity-field abstraction both halves are built on. Section III develops the body skin-flow analysis. Section IV develops the wing attachment line, posed in leading-edge-normal coordinates, and argues that this choice is forced — not merely preferable — once sideslip is admitted. Section V verifies both against closed-form results. This part carries the aerodynamics *up to* the separation boundary: the coefficient and derivative tables it produces are truncated where the march says separation takes over. What lies beyond that boundary — the post-separation coefficients toward the flat-plate limit — is the subject of Part II of this series.

The motivating application is an attitude sweep. Angle of attack and sideslip are the two parameters a stability and control analysis varies most, and they move the attachment line in qualitatively different ways: α walks it chordwise, symmetrically; β moves it *asymmetrically* on a swept wing and rotates the body's stagnation point about the fuselage axis. The second effect has consequences for transition that a symmetric treatment cannot represent.

II. Formulation

A. The coupled wing–body system

Lifting surfaces are discretized into horseshoe-vortex panels with bound segments on the local quarter-chord line and flow tangency enforced at the three-quarter-chord point. The fuselage is discretized into constant-strength quadrilateral source panels: a source distribution enforces zero through-flow on a closed non-lifting volume without introducing bound circulation or a shed wake, which paneling a fuselage with horseshoes would incorrectly do. Sources produce no net force on a closed body, as d'Alembert requires, but they do produce the surface pressure distribution and hence the Munk moment,[6] and the upwash the wing sees.

The two are not solved separately and then superposed. There is one system, in which every unknown is a singularity strength and every equation is flow tangency at a control point:

$$\begin{bmatrix} A_{vv} & A_{vs} \\ A_{sv} & A_{ss} \end{bmatrix} \begin{Bmatrix} \Gamma \\ \sigma \end{Bmatrix} = \begin{Bmatrix} w_v - \mathbf{V}_{\text{kin}} \cdot \hat{\mathbf{n}}_v \\ w_s - \mathbf{V}_{\text{kin}} \cdot \hat{\mathbf{n}}_s \end{Bmatrix} \tag{1}$$

where A_{xy} is the normal velocity induced at an x control point by unit strength on a y element and w is the prescribed normal velocity, zero on any solid surface. The blocking is emergent rather than constructed: unknowns are indexed vortices first, then sources, so the four blocks are regions of one matrix filled by one loop. A wing-only case is the same system with the source rows and columns absent.

In-plane source influence follows the classical Hess–Smith flat-panel result.[7,8] The normal component is taken as the signed solid angle the panel subtends at the field point, evaluated per triangle by the van Oosterom–Strackee formula[9] rather than by the textbook per-edge arctangent form, which is singular for any edge parallel to a local axis — a case that arises constantly on a body of revolution.

B. The velocity field as a first-class object

Both analyses that follow must evaluate velocity *after* the solve, at points that are not control points. This is a modest but consequential change in how a panel solver is structured: the field is made an explicit object carrying the converged Γ and σ, and the force integration is rewritten to go through it, so that no second definition of "the velocity here" exists to drift from the first.

Three evaluations are distinguished, and neither of the special cases is an approximation of the general one. The plain field applies at an arbitrary point. At a panel's own bound-vortex midpoint that segment is singular and is excluded, because Kutta–Joukowski requires the flow the segment sits *in*, not the flow it creates. At a source panel's own control point — which lies exactly on its own sheet, where the induced velocity is indeterminate — the analytic $+\frac{1}{2}$ sheet jump is substituted, the in-plane part canceling by symmetry.

III. Body skin-flow topology

A. Surface coordinates

The analysis is carried out in the paneling's own (i, j) index space — axial station and azimuthal sector — rather than in a body-fixed (x, ϕ). Recovering an azimuth as $\arctan(z/y)$ presumes the surface is swept about a particular axis through the origin, which is true of a fuselage, false of an offset duct or nacelle, and *silently* false: the failure mode is a streamline that curves incorrectly, not an error. Each panel therefore states its own surface indices, and a patch that does not state a complete topology is declined rather than reconstructed.

Index space requires a metric, carried on the grid:

$$\mathbf{a}_i = \frac{\partial \mathbf{P}}{\partial i}, \quad \mathbf{a}_j = \frac{\partial \mathbf{P}}{\partial j}, \quad \mathbf{V}_t = u\,\mathbf{a}_i + v\,\mathbf{a}_j, \quad \sqrt{g} = |\mathbf{a}_i \times \mathbf{a}_j|. \tag{2}$$

The contravariant components (u, v) are obtained once per node from the 2×2 Gram system and stored. Because the Gram matrix is positive definite, they vanish exactly where $\mathbf{V}_t$ does, which reduces the search for stagnation points to the search for a common zero of two bilinear interpolants: cells across whose corners both components change sign are candidates, and a two-dimensional Newton iteration locates the zero within one.

B. Classification

A critical point is classified from the Jacobian of the surface flow. The eigenvalues are coordinate-independent: where the velocity vanishes, a change of surface coordinate maps $J \mapsto MJM^{-1}$, the term involving the derivative of M being multiplied by a velocity that is zero. Node, saddle and focus, and the attachment/separation distinction, are therefore properties of the flow and not of the chart. Surface streamlines run along $\mathbf{V}_t$, so flow *arrives* and spreads at an attachment point and its eigenvalues have positive real parts.

Two numerical points determine whether this is usable in practice.

First, the Jacobian is *not* differenced from (u, v). Those are not smooth fields even where the flow is: on a body of revolution the azimuthal basis vector shrinks with local radius, so v carries a $1/r$ factor that varies by tens of percent across a single cell near a nose. Interpolating it is harmless — it is exact at the nodes, which is all a streamline integrator requires — but differencing it folds the chart's own distortion into the result. Measured on a sphere, whose stagnation node is exactly isotropic, this produces a spurious 2:1 eigenvalue split. The strain rate is instead obtained by differencing the tangential velocity vector in a local orthonormal frame, which makes J a physical strain rate in s^{-1}.

Second, a focus is declared only when the point genuinely spirals. An isotropic node lies exactly on the node/focus boundary, where the discriminant vanishes; a bare sign test on the discriminant reports a sphere's forward stagnation point as a spiral under any perturbation. The implementation requires $|\mathrm{Im}\,\lambda|$ to reach a quarter of $|\mathrm{Re}\,\lambda|$, reserving the classification for the vortical separation it is meant to name.

C. Streamlines and the spreading metric

Surface streamlines are integrated with fourth-order Runge–Kutta in index space, so that a step is a fixed fraction of a *cell* rather than of a length — the natural step for a mesh whose cells vary in size along the body. Near a critical point the field is $\dot{\xi} = \lambda\xi$, so a step scaled as

$1/|\mathbf{V}|$ grows without bound; the step is therefore additionally capped by displacement.

Each traced line carries what a boundary layer consumes. Besides arc length and edge speed, this includes the streamline-spreading metric $h(s)$, integrated from the surface divergence of the unit flow direction,

$$\frac{d\ln h}{ds} = \nabla_s \cdot \hat{\mathbf{e}}, \qquad \hat{\mathbf{e}} = \mathbf{V}_t/|\mathbf{V}_t|, \tag{3}$$

which is what separates the three-dimensional momentum-integral equation from its two-dimensional form,

$$\frac{d\theta}{ds} + (H+2)\frac{\theta}{U_e}\frac{dU_e}{ds} + \frac{\theta}{h}\frac{dh}{ds} = \frac{c_f}{2}. \tag{4}$$

Where streamlines fan apart h grows and the layer is thinned; where they converge h collapses, which is the signature of a separation line.

D. The zero-incidence case

At $\alpha = \beta = 0$ the stagnation point of a body of revolution lies on the nose apex, which is the collapsed forward edge of the first panel ring and not a control point. There is no interior zero to find, and the analysis reports the fact rather than nominating the nearest sample. Any nonzero incidence moves the point onto the surface proper. This is a small matter of interface honesty with a practical consequence: on a slender body at low incidence the stagnation point can lie *within* the first panel, and a method that always returns a located point conceals the need for nose refinement.

IV. The wing attachment line

A. Recovering thickness

The section shapes are carried as class–shape transformation (CST) coefficients,[10] from which the lattice uses only the mean line. The thick closed contour is reconstructed from the same coefficients, cosine-spaced at both ends — the leading-edge clustering is not optional, since the stagnation point lies within a percent or so of chord of the leading edge and uniform spacing would place the entire quantity of interest inside the first panel. The leading-edge radius follows from the parameterization without fitting: with class exponent $N_1 = 1/2$ the surface behaves as $A_0\sqrt{\psi}$ near the nose, giving

$$r_{LE}/c = A_0^2/2. \tag{5}$$

The contour is solved by the classical Hess–Smith construction:[7,11] N constant-strength source panels carrying thickness plus a single constant vortex strength over the whole surface carrying circulation, against N tangency equations and one Kutta condition. Sources alone cannot lift and a distributed vortex alone cannot produce a closed body's thickness; the *shared* vortex strength is what keeps the system square with exactly one Kutta condition.

B. Where the stagnation point lies

Measured along the surface from the leading edge, the stagnation-point offset follows

$$\frac{s_{\text{stag}}}{c} \sim \sqrt{\frac{2r_{LE}}{c}}\,\alpha_e, \tag{6}$$

with α_e the incidence from the zero-lift line. The square root is worth emphasizing because the intuitive reading is wrong by an order of magnitude. A round nose does not behave as an isolated cylinder in a stream inclined at α, which would give $s \sim r_{LE}\alpha$; it sits inside the leading-edge singularity of the outer thin-airfoil solution, where $u \sim \alpha\sqrt{c/s}$. Matching this against the parabolic nose's own surface speed $\sqrt{s/r_{LE}}$ yields (6). For a NACA 0012 at four degrees the cylinder estimate gives $0.001\,c$ against an actual $0.013\,c$. This gap is the argument for solving the section rather than correlating it.

C. Leading-edge-normal coordinates, and why sideslip forces them

The section cuts carried by a geometry definition are streamwise: taken at a span fraction, along the flight direction. This is the wrong plane in which to pose an attachment-line problem once there is sweep. Infinite-swept-wing theory separates the flow near a swept leading edge into a two-dimensional problem in the plane *normal* to it, plus a spanwise velocity that is simply convected. The normal problem places the attachment line; the spanwise velocity does not affect where it is, but is the whole of what decides whether it remains laminar. The streamwise contour maps into the normal plane as

$$\psi_n = \psi, \qquad \zeta_n = \zeta/\cos\Lambda, \qquad r_{LE,n} = r_{LE}/\cos^2\Lambda, \tag{7}$$

once renormalized by the shortened normal chord $c_n = c\cos\Lambda$: the section becomes thicker and its nose blunter.

Sideslip is what makes this structural rather than a refinement. The effective sweep,

$$\Lambda_{\text{eff}} = \arcsin\left(\frac{|\mathbf{V}\cdot\hat{\mathbf{e}}_{LE}|}{|\mathbf{V}|}\right), \tag{8}$$

is not the geometric sweep once $\beta \neq 0$, and it moves in *opposite* directions on the two wings: at positive sideslip one leading edge is swept further from the flow while the other is raked toward it. To first order $\Lambda_{\text{eff}} \approx \Lambda \pm \beta$. The attachment line consequently shifts asymmetrically, and — more importantly — the two wings differ in their proximity to leading-edge contamination. A streamwise formulation returns a symmetric answer to an asymmetric question.

A further detail matters at the root. A swept wing's leading edge is not differentiable at the centerline: it runs aft going outboard on both sides, so its direction is discontinuous there. A central difference of leading-edge position across the centerline returns very nearly an unswept direction, assigning the two innermost strips roughly half the true sweep. The break is detected by comparing one-sided differences, and the side that does not cross it is used. This is not merely a numerical tidiness: the root is precisely where the disturbance that contaminates an attachment line originates.

D. Boundary-layer initial conditions

Locating the attachment line is half of what a coupling requires; the other half is the state in which the march begins, and both emerge from the same solve. The strain rate dU_e/ds at the attachment point sets the chordwise layer's initial momentum thickness through Thwaites' stagnation value,[12]

$$\theta_0^2 = 0.075\,\frac{\nu}{dU_e/ds}, \tag{9}$$

which is finite, and is *not* the zero that a flat-plate start assumes. The error is largest exactly where the layer is thinnest and the transition correlations most sensitive.

The swept attachment line carries its own boundary layer, with length scale and momentum thickness

$$\eta = \sqrt{\nu/(dU_e/ds)}, \qquad \theta_{AL} = 0.404\,\eta, \tag{10}$$

and a state governed by Poll's attachment-line Reynolds number,[13]

$$\bar{R} = \frac{W_e\,\eta}{\nu}, \tag{11}$$

with W_e the velocity along the leading edge. The two thresholds mean different things. Below $\bar{R} \approx 245$ a disturbance introduced at the root — a fuselage junction, a trip, a de-icing boot — decays as it travels outboard. Above it the disturbance propagates and contaminates the entire leading edge, turning the whole wing turbulent regardless of what the chordwise flow would have done.[14,15] Above $\bar{R} \approx 583$ the attachment line becomes turbulent unaided. Because W_e follows the effective sweep, $\bar{R}$ is asymmetric in sideslip: one wing can be contaminated while the other is not, at the same instant on the same aircraft.

E. Marching to separation

Everything above is a prerequisite; this subsection spends it. Each strip's two surface runs are marched with Thwaites' method laminar, Michel's criterion for transition, and Head's entrainment method with Ludwieg–Tillmann skin friction turbulent,[1] starting from the attachment point rather than from the leading edge.

Thwaites supplies his own initial condition when the integral starts at the attachment point, which is worth stating because it looks like a missing input. With $U_e \sim a\,s$ near the stagnation point,

$$\theta^2 = \frac{0.45}{Re\,U_e^6}\int_0^s U_e^5\,ds \;\longrightarrow\; \frac{0.45\,a^5 s^6}{6\,Re\,a^6 s^6} = \frac{0.075}{Re\,a}, \tag{12}$$

which is θ_0 again, reached by an entirely different route. The march needs no seed, and the agreement between the two is a check rather than a coincidence.

One numerical point earns its place here because it is worth a factor of three. The integrand is U_e^5, and on the *first* step out of the attachment point U_e rises linearly from zero, so the trapezoid rule returns $h\,U_{e,1}^5/2$ where the true value is $h\,U_{e,1}^5/6$ — a factor of three in the integral and hence $\sqrt{3}$ in θ_0, the one quantity a march from a stagnation point exists to get right. Integrating the piecewise-linear interpolant exactly, $\int_0^h U_e^5\,ds = h\,(U_{e,1}^6 - U_{e,0}^6)/[6(U_{e,1} - U_{e,0})]$, removes it. A march that starts at a leading edge with $\theta = 0$ never encounters this, having no θ_0 to be wrong about.

What counts as separation requires a decision, and the obvious one is useless. At the Reynolds number of a small airframe ($Re_n \approx 3\times10^5$ here) the laminar layer reaches Thwaites' $\lambda = -0.09$ before Michel's criterion trips at *every* incidence in the matrix of Section VI, including negative ones. Reporting that as separation is true of the laminar layer and draws no boundary: it marks every attitude separated. What occurs physically is a laminar separation *bubble* — the sheared layer transitions immediately downstream and reattaches turbulent within a few percent of chord, which is the defining behavior of low-Reynolds airfoils rather than a failure of the flow. The bubble is therefore treated as the transition trigger, the march continues turbulent, and the verdict is *turbulent* separation, $H \geq 2.4$: the trailing-edge separation point moving forward, which is what stalls a wing. The laminar separation point is still reported, both because it is the part of the march verifiable in closed form and because a designer wants to know where the bubble sits.

The limitation this leaves is bubble bursting. A short bubble that fails to reattach is what actually ends the lift curve of a thin section below $Re \approx 2 \times 10^5$, and predicting it requires a bubble-length correlation or an e^N envelope that this method does not carry. The separation boundary reported in Section VI is consequently an upper bound on usable incidence, not a stall prediction, and is labeled as such.

V. Verification

All three parts are verified against closed-form results rather than against mesh refinement alone.

A. Sphere: exact stagnation points at arbitrary attitude

Potential flow past a sphere has surface velocity

$$\mathbf{V}_t = \tfrac{3}{2}\left(\mathbf{U} - (\mathbf{U}\cdot\hat{\mathbf{n}})\,\hat{\mathbf{n}}\right) \tag{13}$$

exactly, so it vanishes precisely where the outward normal is parallel to the freestream: an attachment point at $\hat{\mathbf{n}} = -\hat{\mathbf{U}}$ and a separation point at $+\hat{\mathbf{U}}$. This holds at *any* α and β, which makes the sphere a complete test of the attitude sweep rather than of a single condition. Both points are recovered to within half a panel across $\alpha \in [-12^\circ, 15^\circ]$, $\beta \in [-6^\circ, 12^\circ]$.

The strain rate is likewise exact: $|\mathbf{V}_t| = \frac{3}{2}Us/a$ near the stagnation point gives both eigenvalues as $\frac{3}{2}U/a$ and $\operatorname{tr} J = 3U/a$. The computed trace converges to within approximately 4% at $N = 48$ per direction. The eigenvalue ratio, which must be unity for this isotropic (star) node, converges as $0.28, 0.67, 0.85, 0.94$ for $N = 24, 32, 48, 64$ — and, notably, does *not* converge when the Jacobian is taken from the raw contravariant components, which is the measurement that motivated the orthonormal-frame formulation of Section III.

Surface streamlines from the stagnation point are meridians, so the spreading metric must satisfy $h \propto \sin\theta$; the computed ratio $h(s_2)/h(s_1)$ matches $\sin\theta_2/\sin\theta_1$ to within 15%.

A prolate spheroid supplies a symmetry statement that no discretization can satisfy accidentally: for any body of revolution about the x axis, the attachment point must lie on the windward meridian, $\phi = \arctan_2(-\sin\alpha\cos\beta,\ -\sin\beta)$. This is recovered to within one azimuthal sector at all tested attitudes, and is the assertion that would detect a sign error in the attitude convention.

B. Cylinder and section: exact surface velocity

Supplied a circle rather than an airfoil, the section solver must reproduce $|V| = 2U\sin\theta$ and $C_p = 1 - 4\sin^2\theta$ with zero circulation — a result owing nothing to airfoil theory, and one that exercises the influence coefficients, assembly and solve together. Worst surface-speed error is below $0.01\,U$ at 240 panels, and the stagnation-point strain rate matches the exact $2U/a$ to within 5%.

A symmetric section at zero incidence stagnates exactly on the leading edge and produces no lift. The lift slope sits just above 2π, as thickness requires. The scaling of (6) is verified as a *collapse*: $s_{\text{stag}}/(\sqrt{r_{LE}}\,\alpha)$ is constant to within 12% across a twelvefold range of nose radius, which no fitted constant can reproduce if the scaling law itself is wrong.

C. The separation march

The march of Section E is pinned against three closed-form answers and one control. A flat plate must recover Blasius: Thwaites gives $\theta = \sqrt{0.45\, s/Re}$, the known 1% above the exact $0.664\, s/\sqrt{Re_s}$, with $H = 2.61$ throughout and no separation. Howarth's linearly retarded flow $U_e = U_0(1 - s/L)$ is the textbook test, and must place laminar separation at $s/L = 0.123$ against the exact 0.120; the computed value must also be converged under mesh refinement, which a crossing quantized to whole stations would fail. A run with $U_e = a\, s$ must produce $\theta_0 = \sqrt{0.075/(Re\, a)}$ unseeded across a sixteenfold range of strain rate — the check that the march begins where it claims to, and the one that exposed the trapezoid error in Eq. (12). The control is an *accelerating* flow at the same Reynolds number and length, which must neither bubble nor separate; without it, the Howarth test would also be passed by an implementation that reported separation eagerly.

D. Sideslip asymmetry

The claim of Section IV is tested directly. A 30° swept wing at $\beta = 10°$ must show effective sweeps near $\Lambda + \beta$ and $\Lambda - \beta$ on the two wings, with the attachment-line Reynolds number following; reversing the sideslip must exchange them exactly. Both hold. The necessary control is an *unswept* wing at the same sideslip, which must remain symmetric — without it, the swept test would also be passed by an implementation that merely keyed on the sign of the spanwise coordinate. The root kink is detected at exactly the two strips flanking the centerline, and a straight leading edge is not reported as kinked.

VI. Application

The method is exercised on the solver's own regression airframe: a VBAT-class tail-sitter described by a schema-1.8.0 geometry handoff, carrying a rectangular wing of span $b = 1.063$ m and chord 0.177 m ($AR = 6.0$) with *zero* sweep, twist and dihedral, a body of revolution of length $L = 0.491$ m and maximum radius 48.8 mm, and an aft duct. The planform is worth emphasizing before the results: every station of this wing has a geometric sweep of exactly zero, so every degree of effective sweep reported below is put there by the flow field, not by the drawing. Conditions are $V_\infty = 25$ m/s at sea level, $\alpha \in \{0, 4, 8\}°$ and $\beta \in \{-10, 0, +10\}°$.

The lattice carries one Weissinger row per strip (44 strips), which is the contract the attachment-line analysis states: one local velocity per spanwise station, read at the bound-

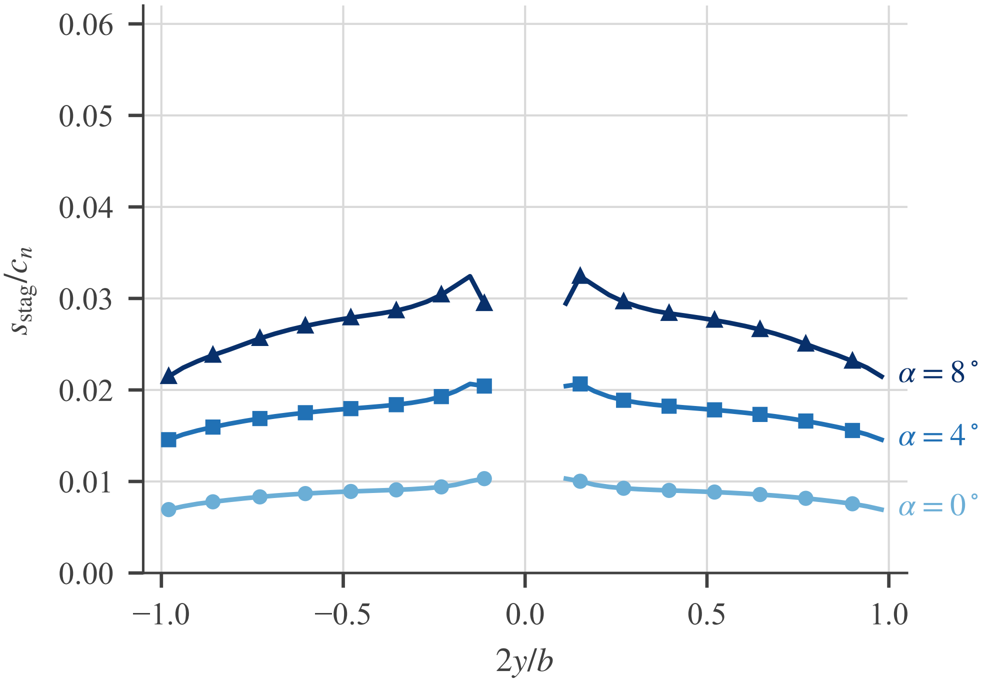

Figure 1. Attachment-line position across the span at $\beta = 0$: surface offset of the stagnation point from the leading edge, as a fraction of the normal chord, positive onto the lower surface. The curves break at the fuselage. The offset is nonzero at $\alpha = 0$ because the section is cambered.

vortex midpoint. The fuselage carries 768 source panels and the duct 160. Two discretization choices deserve a sentence each, because both are consumer-side choices the handoff deliberately leaves open. First, the body's station list is resampled with cosine clustering toward the nose, keeping every original breakpoint so the added stations lie exactly on the contract's piecewise-linear radius law: at these incidences the stagnation point sits a few millimetres from the apex, inside the first panel ring of the as-delivered spacing, where the honest answer of Section III is "attaches upstream" rather than a located node. Second, the wing is trimmed at the body surface ($2y/b = 0.092$) and the root circulation is carried through the fuselage by extending the innermost bound segments to the centreline. That carry-through segment invalidates its own strip's leading-edge geometry, so the attachment line is evaluated on a companion solve that is identical except for the carry-through; the sensitivity this introduces is quantified below rather than assumed away.

A. Incidence sweep

Figure 1 shows the attachment line at $\beta = 0$. Every exposed station resolves a stagnation point at every condition, and the mean offset grows from $0.009\,c_n$ through $0.018\,c_n$ to $0.027\,c_n$ across the α sweep, tapering toward the tips as induced downwash relieves the local incidence (the normal-plane incidence falls from 6.4° near the root to 3.3° at the tip at $\alpha = 8°$). The nonzero offset at $\alpha = 0$ is not noise: a cambered section at zero incidence stagnates below its leading edge, at $0.009\,c_n$ here — which is precisely the initial condition a boundary-layer march loses when it starts from the camber line's leading edge with $\theta = 0$, the approximation Section IV exists to remove. Lift coefficients for the coupled airframe are 0.327, 0.663 and 0.992 across the sweep.

B. Sideslip and the root junction

Figure 2 contains the two results of this section. The first concerns the outboard wing. Infinite-swept-wing reasoning — and the isolated-wing control case of Section V — says an unswept wing at sideslip is left/right symmetric, with $\Lambda_{\text{eff}} = |\beta|$ on both sides. The coupled airframe disagrees: at $\beta = +10°$ the outboard plateau reads 10.3° on the windward wing against 9.4° on the leeward at $\alpha = 8°$, and the split shrinks with lift (10.1 against 9.6 at 4°, 9.9 against 9.7 at 0°) while reversing exactly under $\beta \to -\beta$. On a planform with identically zero sweep, that asymmetry is supplied entirely by the induced field — the trailing-vortex system and the body's sidewash, neither of which an isolated-section treatment sees.

The second result is the root junction. At $\beta = 0$ the effective sweep climbs from under 1° over the outer wing to 30° at the station beside the fuselage, and $\bar{R}$ climbs with it, from below 2 to 63 at $\alpha = 8°$: the body's crossflow at incidence must pass around the wing root, and the spanwise flow it acquires in doing so is what a swept leading edge would otherwise have to supply. The attachment-line boundary layer of an unswept wing–body junction is, in this specific sense, all junction and no wing. Sideslip then splits the junction: the windward root sharpens to 32° and $\bar{R} = 66$, while one station out on the leeward side the spanwise flow passes through a null ($\Lambda_{\text{eff}} = 0.9°$, $\bar{R} < 2$) where the body's sidewash cancels the sideslip component, before recovering to the outboard plateau. Since leading-edge contamination is seeded at exactly this junction — it is Poll's root disturbance[13] — the coupled solve's verdict is that the two roots are not equally exposed — and its engineering summary for this airframe is unambiguous: the worst $\bar{R}$ anywhere in the attitude matrix is 135, a factor 1.8 below the contamination threshold, so the attachment line stays laminar with margin at every station and attitude examined.

The junction numbers carry a stated modelling sensitivity. Repeating the sweep with the carry-through vortex enabled moves the outboard stations almost nowhere — offsets by less than $0.002\,c_n$ — but moves the two stations flanking the fuselage by up to 13° of effective sweep and by up to the full magnitude of their $\bar{R}$. The root spike and its sideslip split are robust; their third significant figure is not, and the figures should be read accordingly. A practical meshing note belongs with this: the trimmed wing's inner trailing legs run along the body flank at exactly the trim radius, and an azimuthal body refinement that walks

Table 1. The separation boundary: forwardmost upper-surface separation, with the percentage of span separated forward of $x/c_n = 0.90$ in parentheses. A dash means no station separates forward of that limit.

α	$\beta = -10°$	$\beta = -5°$	$\beta = 0°$	$\beta = 5°$	$\beta = 10°$
-4	—	—	—	—	—
-2	—	—	—	—	—
0	—	—	—	—	—
2	—	—	—	—	—
4	—	—	—	—	—
6	—	—	—	—	—
8	—	—	—	—	—
10	—	—	—	—	—
12	0.88 (55%)	0.88 (55%)	0.88 (55%)	0.88 (55%)	0.88 (55%)
14	0.84 (82%)	0.84 (82%)	0.85 (82%)	0.84 (82%)	0.84 (82%)
16	0.79 (98%)	0.80 (98%)	0.80 (100%)	0.80 (98%)	0.79 (98%)

flank control points onto that line degrades the system's pivot ratio by a factor of five and inflates the solution. The sector count here (16) keeps them 9.6 mm clear; the criterion, as always with a lattice, is distance between singularities and control points, not panel count.

C. Fuselage stagnation point across the sweep

Figure 3 tracks the body's primary attachment node. At $\alpha = \beta = 0$ the flow is axisymmetric and the stagnation point is the apex itself, upstream of every surface sample — the case Section III declines to invent a node for. Every other condition resolves an interior node, and its migration is the body-side picture of what the attitude sweep does: pure α walks it aft along the windward meridian ($x/L = 0.0006$ to 0.0021 from $4°$ to $8°$); adding sideslip rotates it off the symmetry plane to $86°$, $62°$ and $45°$ from windward as α grows against a fixed $\beta = 10°$, the node sliding around the blunt nose toward the resultant of the two incidences while also moving aft ($x/L = 0.0049$ at the steepest condition). Reversing β mirrors every position exactly. This node is the seed of every boundary-layer march on the body, and its strain rate — evaluated in the local orthonormal frame of Section III, not from the chart components — sets θ_0 there just as the section strain rate does on the wing.

D. Where the flow separates

The march of Section E is run at every attitude, and Fig. 4 and Table 1 report it. Below $\alpha = 10°$ the turbulent layer holds close to the trailing edge at every station: separation exists, as it always does, but sits aft of $x/c_n = 0.90$ where it is a trailing-edge detail rather than a limit. At $\alpha = 12°$ it moves forward of $x/c_n = 0.90$ over half the span (55%), at 14°

over 82%, and by 16° it covers essentially the whole span with the forwardmost station at $x/c_n \approx 0.80$. The boundary this draws is $\alpha \approx 12°$, and it is the same at every sideslip angle tested.

The spanwise pattern is worth reading rather than summarizing. Separation is consistently most advanced just *outboard of the wing root*, near $2y/b \approx 0.2$, and recovers monotonically toward the tips — at $\alpha = 16°$, from $x/c_n \approx 0.80$ inboard to 0.88 at the tip. That ordering follows from the same induced field that flattens the attachment line's offset outboard: downwash relieves the local incidence toward the tips, so the outer sections run a milder pressure recovery than the inner ones and hold their layers longer. The practical reading is that this planform sheds its root first, which is the benign direction — the ailerons stay in attached flow past the attitude at which the inboard wing has let go.

That insensitivity is worth stating explicitly, because it is the opposite of what the attachment line does. Sideslip moves the attachment line and its Reynolds number substantially and asymmetrically (Fig. 2), yet leaves the separation boundary alone. The two are not in conflict: the attachment line responds to the *spanwise* component of the flow, which is exactly what sideslip changes, while separation is driven by the chordwise pressure recovery in the leading-edge-normal plane, which β leaves nearly untouched on an unswept wing. A method that reported the attachment line alone would suggest a strong sideslip sensitivity that the separation behaviour does not share, and vice versa. Both are needed to say which one an attitude limit should be drawn from — here, separation.

Transition is by laminar separation — a bubble — rather than by Michel's criterion at 2386 of the 2420 station-attitudes in the matrix (98.6%; the 34 exceptions cluster at $\alpha = 8$–12°). That is the expected behaviour at $Re_n \approx 3 \times 10^5$, and it is the reason the verdict is drawn from turbulent separation rather than from $\lambda = -0.09$ (Section E). The caveat stated there applies to the whole of this subsection: bursting is not modelled, so $\alpha \approx 12°$ is an upper bound on the attitude at which the potential-flow coefficients can be trusted, not a predicted stall.

E. Coefficients and stability derivatives

Tables 2 and 3 are the practical output of the whole exercise: a coefficient and derivative set over the attitude matrix, restricted to the attitudes at which Section D says the potential-flow answer is still meaningful. Restricting them is the point. A panel method will return a lift coefficient at any incidence it is asked for, and the returned number at $\alpha = 16°$ ($C_L = 1.62$) is arithmetically sound and physically meaningless with most of the upper surface separated. Having the separation march in the same solve is what lets the table be truncated on evidence instead of on a rule of thumb.

The derivatives are computed on the *coupled* system, not on the wing alone. That

Table 2. Force and moment coefficients over the attached portion of the attitude matrix. Rows at $\alpha \geq 12°$ are omitted: the separation march of Section D places separation forward of $x/c_n = 0.90$ there, and a potential-flow coefficient past that point is an extrapolation rather than a prediction. C_{D_i} comes from a near-field integration and carries a spurious body contribution — it is negative at zero lift and is discussed, rather than relied on, in the text. The aerodynamics *beyond* this table's edge — the post-separation coefficients toward the flat-plate limit — is the subject of Part II.

α	β	C_L	C_{D_i}(near)	C_{D_i}(far)	C_Y	C_m	C_l	C_n
-4	0	-0.0135	-0.00617	0.00007	0.00000	-0.0097	0.00000	0.00000
-4	10	-0.0140	-0.00448	0.00009	0.00128	-0.0087	-0.00056	-0.00467
-2	0	0.1573	-0.00420	0.00123	0.00000	-0.0138	0.00000	0.00000
-2	10	0.1515	-0.00263	0.00122	0.00092	-0.0127	-0.00050	-0.00468
0	0	0.3273	-0.00035	0.00488	0.00000	-0.0182	0.00000	0.00000
0	10	0.3164	0.00100	0.00475	0.00025	-0.0169	-0.00045	-0.00469
2	0	0.4961	0.00530	0.01098	0.00000	-0.0228	0.00000	0.00000
2	10	0.4801	0.00635	0.01067	-0.00073	-0.0214	-0.00039	-0.00470
4	0	0.6635	0.01265	0.01952	0.00000	-0.0277	0.00000	0.00000
4	10	0.6423	0.01333	0.01895	-0.00199	-0.0261	-0.00034	-0.00469
6	0	0.8289	0.02159	0.03044	-0.00000	-0.0327	0.00000	0.00000
6	10	0.8027	0.02182	0.02955	-0.00352	-0.0310	-0.00028	-0.00468
8	0	0.9920	0.03198	0.04370	0.00000	-0.0380	0.00000	0.00000
8	10	0.9608	0.03170	0.04241	-0.00530	-0.0361	-0.00023	-0.00467
10	0	1.1525	0.04368	0.05924	0.00000	-0.0434	0.00000	0.00000
10	10	1.1164	0.04282	0.05747	-0.00729	-0.0414	-0.00017	-0.00465

matters most for C_{n_β}, which is negative across the whole matrix — this configuration is directionally unstable, as a wing–body with no vertical surface must be, since the fuselage's Munk moment is destabilizing and there is nothing aft to oppose it. A wing-only derivative calculation would report a value near zero and miss the conclusion entirely. C_{m_α} is likewise small and negative, the modest static margin that follows from a moment reference point slightly ahead of the wing's aerodynamic centre, and it is only correct because the reference point is carried through the contract-to-solver frame rotation; skipping that flip places it an equal distance the wrong side of the origin and inflates C_{m_α} by an order of magnitude through a spurious moment arm of about one body length.

Roll damping is the derivative worth checking a method against, because it has a textbook value: C_{l_p} comes out at -0.458 at $\alpha = 0$ and drifts only to -0.440 by $16°$, against the standard result near -0.45 for an unswept wing of this aspect ratio.

C_{D_i} is reported twice in Table 2, and the reason is worth setting out because the near-field column is unusable on this configuration in a way that is easy to miss.

Forces are ordinarily integrated in the NEAR field here, by applying Kutta–Joukowski

Table 3. Stability derivatives at $\beta = 0$ over the attached attitudes, computed by central differences on the coupled wing–body–duct system. Angle derivatives are per radian; rate derivatives are per unit reduced rate ($q\bar{c}/2V$, $pb/2V$, $rb/2V$).

α	C_{L_α}	C_{m_α}	C_{Y_β}	C_{l_β}	C_{n_β}	C_{m_q}	C_{l_p}	C_{n_r}
-4	4.899	-0.113	0.0090	-0.0032	-0.0273	-0.190	-0.4579	0.0008
-2	4.883	-0.121	0.0068	-0.0029	-0.0274	-0.193	-0.4586	0.0011
0	4.856	-0.129	0.0028	-0.0026	-0.0274	-0.197	-0.4588	0.0014
2	4.817	-0.136	-0.0030	-0.0023	-0.0275	-0.200	-0.4584	0.0017
4	4.767	-0.142	-0.0106	-0.0020	-0.0274	-0.202	-0.4574	0.0020
6	4.707	-0.148	-0.0197	-0.0017	-0.0274	-0.205	-0.4559	0.0023
8	4.637	-0.153	-0.0303	-0.0013	-0.0273	-0.207	-0.4539	0.0026
10	4.557	-0.157	-0.0422	-0.0010	-0.0272	-0.209	-0.4512	0.0029

at each bound-vortex midpoint. On a wing alone that is sound. On a coupled configuration it is not: the near-field C_{D_i} is *negative* at $\alpha = -4°$, where the lift is essentially zero and induced drag must vanish, and fitting the $\beta = 0$ column to $C_{D_i} = C_{D_i,0} + kC_L^2$ returns $C_{D_i,0} = -0.0037$ with a k implying a span efficiency of 1.53 — not merely optimistic but impossible, since $e \leq 1$ for any planar wing.

The cause is worth stating precisely, because the tempting explanation is wrong. It is *not* that the discretized body carries a spurious force: closed bodies in this solver satisfy d'Alembert to machine precision in uniform flow, $|F|/(qA) \sim 10^{-16}$, and the force the body carries in the coupled solve is physical — it sits in the wing's upwash and accounts for about 8% of the configuration's lift. The cause is that near-field induced drag is a small difference of much larger quantities. It is the streamwise component of forces dominated by lift, so its relative error is amplified by the lift-to-drag ratio and converges slowly for that reason alone. Introducing a body substantially alters the induced velocity at the wing's bound vortices, most of all near the root, and adds a pressure integration over the body in a strongly non-uniform field. Both errors are negligible beside lift — which is why C_L and the moments are unaffected, consistent with the lift-slope and roll-damping checks above — and comparable with induced drag, which is a far smaller number.

The far-field column removes both symptoms, and it is worth setting out why it is *structurally* immune rather than merely better conditioned.

Take a control volume closed by a plane far downstream — the Trefftz plane — and let ϕ be the perturbation potential in it. All the disturbance the configuration leaves behind is the wake's crossflow kinetic energy, so the induced drag is

$$D_i = \frac{\rho}{2} \iint_T |\nabla_2 \phi|^2 \, \mathrm{d}y \, \mathrm{d}z, \tag{14}$$

with ∇_2 the in-plane gradient. Far downstream the streamwise perturbation has decayed and ϕ satisfies the two-dimensional Laplace equation in the plane, so Green's first identity converts Eq. (14) into a contour integral around the wake trace C,

$$D_i = -\frac{\rho}{2} \oint_C \Delta\phi \, \frac{\partial \phi}{\partial n} \, \mathrm{d}\ell, \tag{15}$$

the outer boundary contributing nothing because ϕ decays like a two-dimensional doublet. On a lifting surface's wake the potential jump is the bound circulation, $\Delta\phi = \Gamma(y)$, and the normal derivative is the downwash, giving

$$D_i = -\frac{\rho}{2} \int_{-b/2}^{b/2} \Gamma(y) \, w_T(y) \, \mathrm{d}y. \tag{16}$$

The factor of one half in Eq. (16) is often mistaken for a convention. It is not. w_T is the downwash induced *in the Trefftz plane*, where each trailing filament runs from $-\infty$ to $+\infty$ and acts as a full two-dimensional vortex; at the lifting line the same filament is semi-infinite and induces half as much. Substituting the lifting-line downwash into Eq. (16) therefore halves the drag while leaving every trend intact — an error that survives a sign check and a convergence study alike.

The structural point follows from Eq. (15) directly. A closed non-lifting body carries a source distribution, not a vortex sheet. Its potential is single-valued, so $\Delta\phi = 0$ on any trace it leaves, and it contributes nothing to the contour integral *regardless of how it is discretized.* That is why the far-field result cannot inherit the near-field evaluation's error: the body is not merely small in Eq. (15), it is absent from it.

Two approximations remain and should be named. The plane is taken normal to the x axis rather than to the freestream, which is consistent with this solver's frozen wake — the vorticity the plane is meant to intercept runs along x — and carries the same $O(\alpha^2)$ error the frozen wake already does. And Eq. (16) presumes the sheet reaches the plane without rolling up. Roll-up redistributes $\Gamma(y)$ but conserves the impulse that sets D_i, so the leading-order result is unaffected.[8]

With this in place the zero-lift intercept falls from -0.0037 to -0.0003, and C_{D_i} is positive at every attitude in the matrix.

Span efficiency then requires one care in its formation, which is worth stating because getting it wrong looks exactly like a defect in the integral. The far-field drag belongs to the *lifting system*, since only the lifting system sheds a wake, so it must be compared against the lifting system's lift. On this airframe the fuselage carries about 9% of the total, and using the configuration C_L returns $e = 1.17$ — still above the bound — while the wing's own lift returns $e = 0.965$, with per-condition values between 0.96 and 0.99 across the attached range. That is the physically admissible answer, and it is the one the method was always producing; only the comparison was mismatched.

C_{m_q} deserves a caveat rather than a comparison. It is reported as roughly -0.20, and that value is a floor rather than an estimate. The lattice carries one chordwise row per strip, so every bound vortex lies on the quarter-chord line and every control point a half chord behind it: a pitch rate about a point *on* that line adds the same upwash at every control point, a uniform change in effective incidence that moves C_L but exerts no moment about the line it acts on. Pitch damping here therefore arises entirely from the offset between the moment reference point and the quarter-chord line, and the chordwise redistribution of load that supplies most of a real wing's C_{m_q} is absent by construction. Reporting the number with that limitation attached is more useful than either omitting it or presenting it as converged; a chordwise-resolved lattice would be required to do better, at a cost the rest of this method does not otherwise incur.

F. Parasite drag and the complete polar

Everything above concerns C_{D_i}, and a reader entitled to a drag polar is entitled to know what is missing from it. A potential-flow method cannot produce parasite drag at all: a source-panelled closed body carries zero net force in uniform flow to machine precision, which is d'Alembert's paradox behaving correctly rather than a defect of the implementation. The lattice therefore returns the induced component and nothing else, and the complete drag is

$$C_D = C_{D_i} + C_{D_0}, \tag{17}$$

with C_{D_0} supplied by a component buildup of the kind standard in conceptual design:[16,17] skin friction at each component's own Reynolds number, scaled by a form factor for its thickness or fineness and an interference factor for its junction, summed over wetted area.

Evaluated for this airframe at the matrix condition ($V_\infty = 25$ m/s, sea-level ISA) the buildup gives the entries of Table 4. The body and the duct contribute almost equally, which is not obvious in advance: the duct's wetted area is smaller, but its chord is short enough that its Reynolds number is a factor of five lower and its skin-friction coefficient correspondingly higher.

The consequence for the polar is largest where it is least often remarked. At the top of the attached range the induced component dominates and a C_{D_0} of 0.0106 is a modest correction; but at low lift it is the whole of the drag, and a polar plotted from C_{D_i} alone passes through the origin, which no real polar does. The zero-lift intercept discussed in Section E is therefore not merely a numerical artifact to be driven to zero — once Eq. (17) is applied, the physical intercept is C_{D_0}, and the far-field integration's residual -0.0003 is

Table 4. Component buildup of parasite drag at $V_\infty = 25$ m/s, referred to the wing reference area $S = 0.1883$ m^2. The wing is deliberately absent: this table supplies only what the lattice cannot, and the wing's own profile drag belongs to the sectional treatment of Part II rather than to a flat-plate buildup.

Component	S_{wet} [m^2]	Re	C_f	FF	C_{D_0}
Body	0.1422	8.41×10^5	0.00462	1.493	0.00521
Duct	0.1246	1.56×10^5	0.00649	1.181	0.00507
Total, including a 3% excrescence allowance					0.01058

correctly read as small against it rather than against zero.

Two limitations are stated rather than absorbed. The buildup assumes attached flow over both components, so it is valid across this paper's attached matrix and not beyond it; the incidence-dependent separated drag of a body of revolution grows as the cube of $\sin\alpha$ and is the subject of Part II, where the attitude range makes it dominant.[18,19] And a component buildup is a conceptual-design estimate, with a conventional accuracy of $\pm$10–20% for a clean configuration; it is reported here to complete the polar, not as a validated prediction.

G. The aft fan, and what it does to the attached envelope

Everything above is power-off, and on this configuration that is a choice worth defending rather than assuming. The aircraft carries a ducted fan *aft* of the wing—the duct leading edge lies about a fifth of a chord behind the wing trailing edge—so the fan does not blow the wing. A reader may therefore reasonably expect the power-off envelope to be the envelope. It is not quite, and the reason is worth recording in the paper that defines the envelope.

A loaded disk induces flow ahead of itself. The axial induction rises from zero far upstream to v_i in the disk plane, so the wing sits in an *accelerating* stream: a favourable pressure gradient, which is precisely the condition that delays separation. The geometry makes this more than incidental here—the disk radius is 0.19 of the semi-span, against a worst separation station at $|2y/b| \approx 0.15$–0.23 (Section D). The fan sits over the span that sheds first.

The effect is not computable from a momentum-theory slipstream, which returns identically zero ahead of the disk by construction and would report exactly zero interaction—an artifact indistinguishable from a physical null. A uniformly loaded actuator disk is, however, exactly equivalent to a semi-infinite cylindrical vortex sheet, whose field is computable everywhere including upstream. Treating the ducted fan as the annulus it is (two superposed cylinders, shedding at both edges) and adding that field to the coupled solve gives, over the attached range of this paper's matrix,

Two things follow. The increment is real but modest in attached flow—a few percent, growing with thrust and only weakly with incidence—so the attached envelope reported here

Table 5. Fan-induced lift increment over the attached range, as a percentage of the power-off C_L, against thrust coefficient $T_c = T/\bar{q}S$. Both the powered and power-off solves converge to residuals below 10^{-4} throughout this range.

α [deg]	$T_c = 1$	$T_c = 2$	$T_c = 4$	$T_c = 8$
0	2.4%	3.8%	5.9%	8.9%
4	1.7%	2.7%	4.2%	6.4%
8	1.5%	2.4%	3.7%	5.5%
12	1.5%	2.3%	3.6%	5.4%

is not materially displaced by power, and the coefficients of Section E stand as stated. And the increment is largest at low incidence, where the fan's induced velocity is the largest fraction of a lightly loaded stream, rather than near the separation boundary where a delay would matter most.

Whether the fan moves the separation *boundary*—the quantity this paper exists to locate—is not answered by a lift increment, and is deliberately not claimed here. The apparatus to answer it directly is already in place: the march of Section E computes a separation point on every strip, so the powered and power-off separation locations can be compared rather than inferred. That measurement, and the post-stall behaviour where the interaction grows several-fold, belong to Part II and to the fan-induction study respectively.

H. The skin-flow portrait

Figure 5 assembles the whole airframe at the steepest combined condition, $\alpha = 8°$ with $\beta = +10°$. On the wing, the spanwise circulation distribution and the attachment line are shown together, which is the pairing this method exists to produce: the lattice supplies the loading and the local flow, and the section solve turns that local flow into the line the boundary layer starts from, drawn here in the three-dimensional positions the analysis reports, running just below each leading edge from root to tip. On the body, the distinguished streamlines of the attachment node fan out of the windward-left flank of the nose, wrap the body, and collect into a single bundle on the leeward side — rotated well off the geometric top of the body, as the combined incidence demands. The convergence of that bundle is the signature the spreading metric h of Section III exists to carry: streamlines crowding together is precisely h collapsing, which is what an integral method needs to know approaching separation. What the inviscid field does *not* contain is an interior separation critical point — the surface flow leaves the patch through the base annulus, which on this airframe is a duct exit rather than a closed tail, and no zero of the tangential velocity exists forward of it at any condition examined. That is the honest division of labor: an attached potential flow has no business declaring separation, and the topology hands the boundary layer everything it needs — the node to start from, the streamlines to march along, $U_e(s)$ and $h(s)$ down each one — to declare it itself.

VII. Conclusions

A coupled wing–body panel method can supply everything an integral boundary-layer method needs to begin — attachment location, edge velocity distribution, initial momentum thickness, and streamline spreading — provided the wing and the body are treated according to what they actually are. The body's stagnation points are found by searching a real surface field; the wing's cannot be, and must come from a two-dimensional solve of the reconstructed thick section driven by the lattice's induced flow.

Spending those initial conditions on a march to separation turns the method into something a configuration study can use directly: a coefficient and derivative table over an attitude matrix, bounded by evidence rather than by convention. A panel method returns a lift coefficient at any incidence it is asked for, and the burden of knowing which of those numbers to believe has conventionally fallen outside the method. Carrying the boundary layer's starting state through to a separation criterion moves that judgement inside it.

Three formulation choices proved to be load-bearing rather than discretionary. Surface strain rates must be differenced from the tangential velocity in a local orthonormal frame, not from contravariant components whose chart factors do not converge away. The wing's section problem must be posed in the plane normal to the leading edge, not streamwise, because only the former represents the asymmetry that sideslip imposes on a swept wing's attachment line — an asymmetry that determines which wing is closer to leading-edge contamination, and therefore cannot be treated as a second-order correction. And Thwaites' integral must be evaluated exactly on the first step out of the attachment point, where the trapezoid rule is wrong by a factor of three and θ_0 by $\sqrt{3}$ — an error invisible to a march that starts at a leading edge with $\theta = 0$, and therefore one that only appears once the attachment point is located properly.

The results also separate two sensitivities that are easily conflated. Sideslip moves the attachment line strongly and asymmetrically while leaving the separation boundary essentially unchanged, because the two respond to different components of the same flow: the attachment line to the spanwise component, separation to the chordwise pressure recovery. Neither quantity alone would have revealed that, and a configuration limit drawn from the wrong one would be wrong in a way that looks reasonable.

Everything here ends at the separation boundary, deliberately: the coefficient tables stop where the march says the potential-flow answer stops being a prediction. Part II of this series takes up what lies beyond — the post-separation coefficients, anchored on the computed separation points and on the classical flat-plate limits, out to 90° of incidence.

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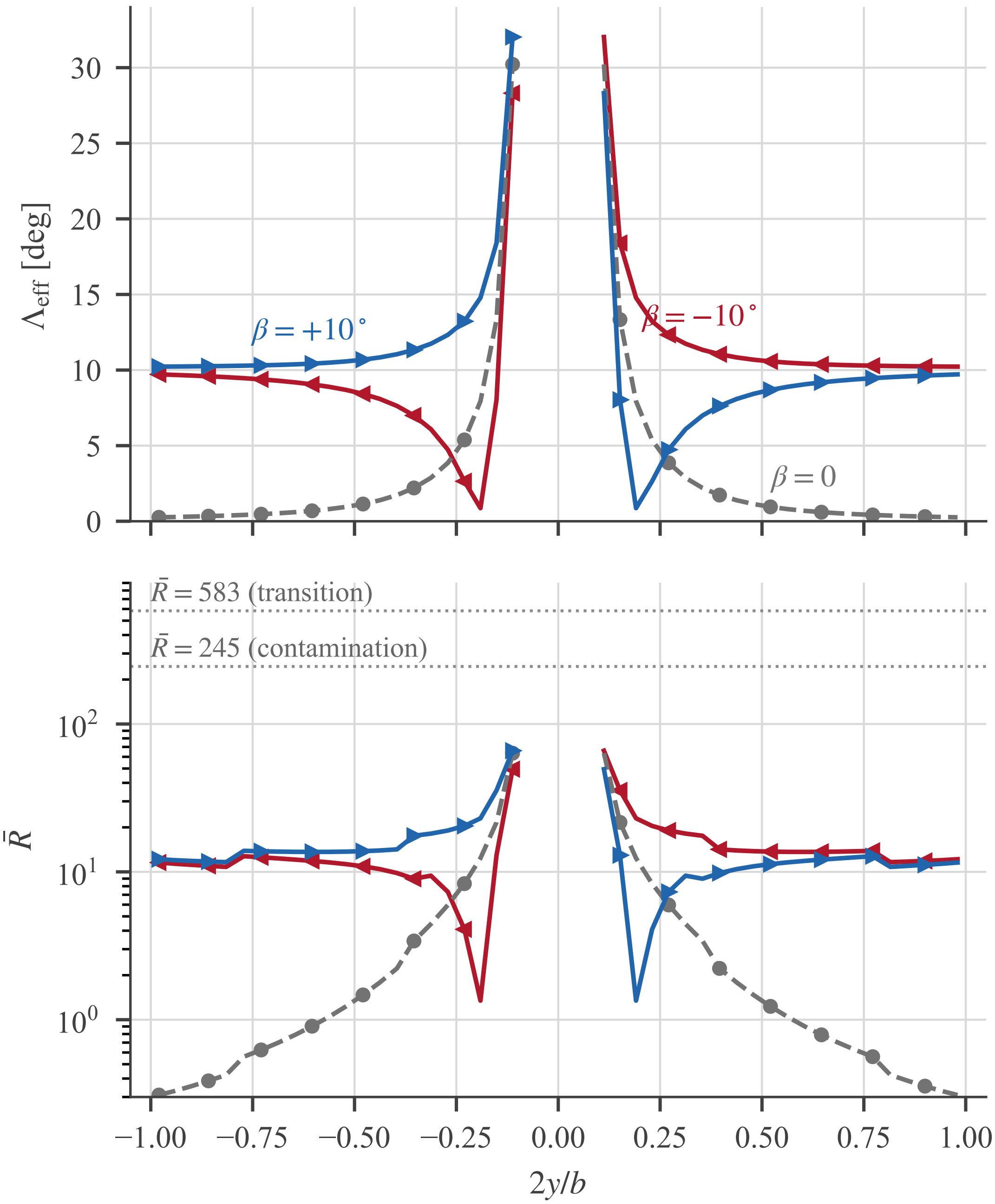

Figure 2. Effective sweep (top) and attachment-line Reynolds number (bottom, log scale) across the span at $\alpha = 8^\circ$, for $\beta = -10, 0, +10^\circ$. Positive sideslip blows toward $+y$, so the left semi-span ($2y/b < 0$) is windward at $\beta = +10^\circ$. Both thresholds of Section IV are shown; the whole airframe stays below them.

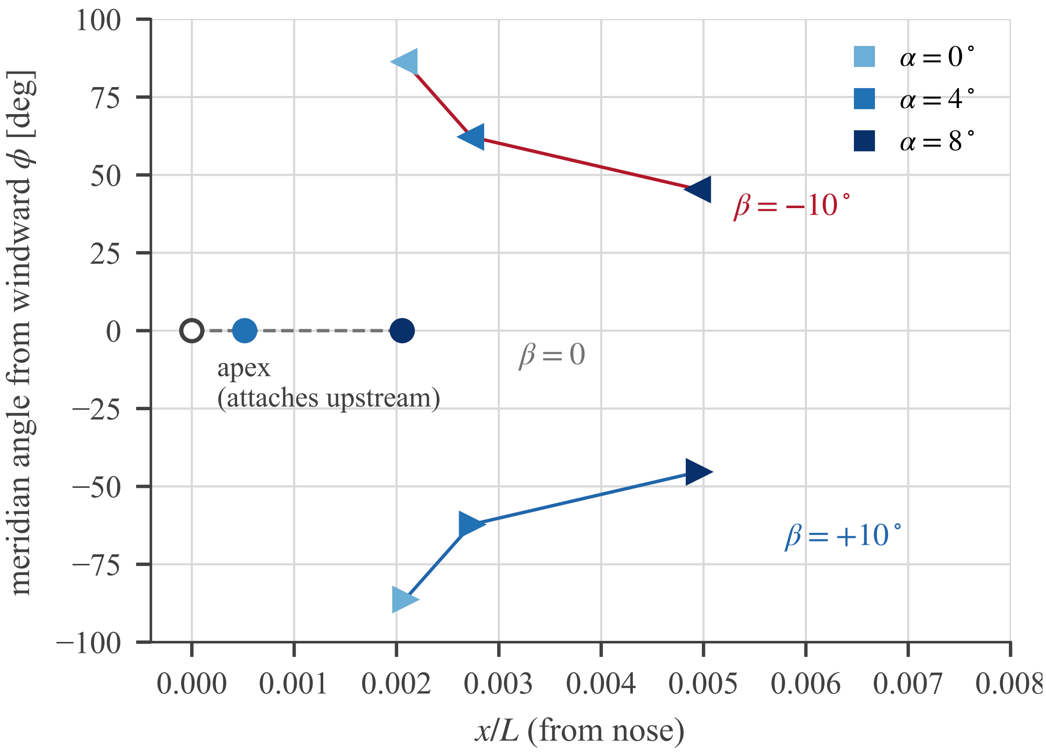

Figure 3. The fuselage attachment node across the sweep, in axial position and meridian angle measured from the windward symmetry plane. Marker shade encodes α; the $\alpha = \beta = 0$ case attaches upstream of the first panel ring and is shown open at the apex.

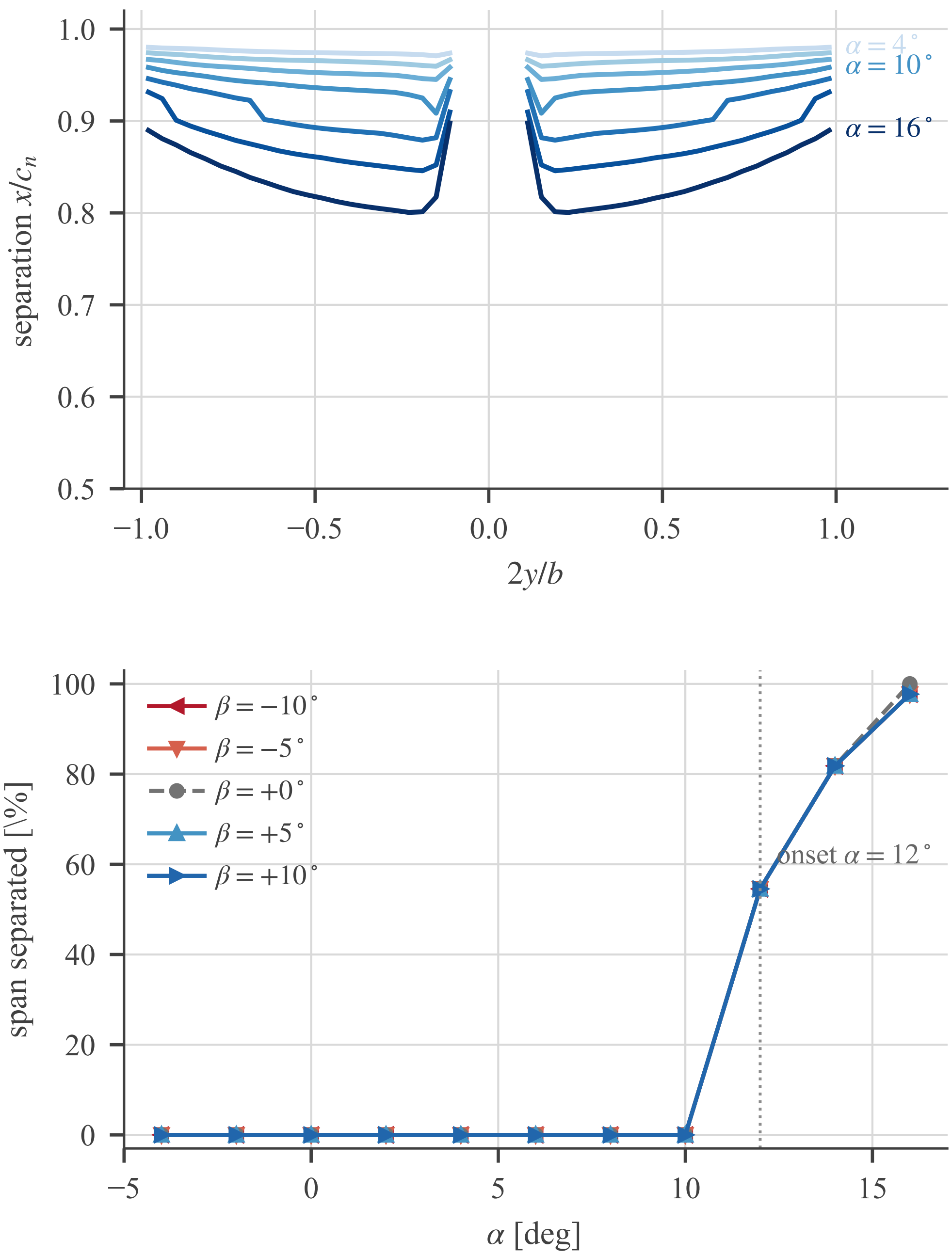

Figure 4. Upper-surface separation from the march of Section E. Top: separation station across the span at $\beta = 0$ for increasing incidence; separation is most advanced just outboard of the wing root and recovers toward the tips. Bottom: fraction of the span separated forward of $x/c_n = 0.90$ against α, for all five sideslip angles — the curves *collapse* — on this unswept planform the separation boundary is insensitive to sideslip.

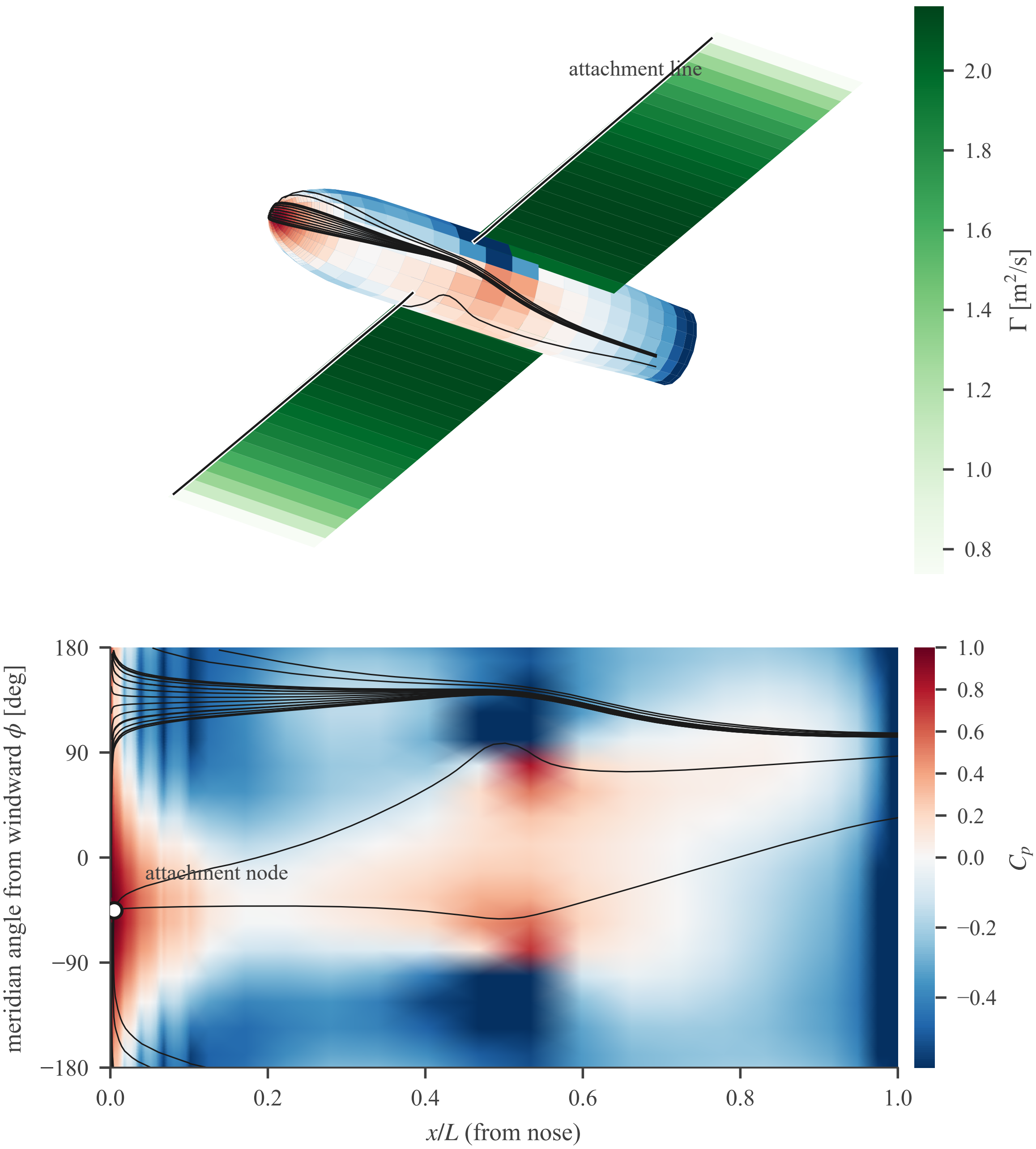

Figure 5. The airframe at $\alpha = 8°$, $\beta = +10°$: orthographic view from the windward-left quarter (top) and the unwrapped fuselage meridian map (bottom). The fuselage carries its surface streamlines colored by C_p referenced to the translational freestream; the wing lattice is colored by its solved bound circulation, with the computed attachment line drawn on the lower surface it lies on. Body streamlines emanate from the attachment node on the windward-left flank and converge into a single leeward line; the duct-exit base annulus is the open downstream boundary.